

Banking Shield

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Banking Shield: Metadata-Only Integrity Evidence and a Deterministic AI Privacy Firewall for a Banking-Adjacent Research Prototype

Version 1.1 — B5-R preprint candidate, 2026-06-13. Full paper produced by Stage B5-D from draft v0.1, the reviewer-simulation fix list, citation verification, final paper-readiness audit, and Stage B5-R independent-style model-assisted hostile review. Academic citations carry DOIs; standards and industry scheme citations use stable institutional URLs. All empirical facts trace to the Stage B5 evidence pack (repo `main` @ `92dabb4`) and were re-audited against checkout `3dcf21b` on 2026-06-13 before these paper-only edits. This paper makes no fraud-detection, scam-prevention, payment-safety, payee-verification, financial-advice, compliance, or production-readiness claims. Substantial LLM assistance was used in drafting and is disclosed in §11.

Preprint status. This manuscript is an author-prepared preprint supported by automated test, audit, claim-review, and model-assisted adversarial-review evidence. It has not yet undergone formal peer review, external banking review, or independent security validation.

Abstract

Research prototypes that touch banking-adjacent workflows face a double bind: they must demonstrate integrity to participants, yet any data they collect — and any AI-generated explanation they emit — creates privacy and overclaim risk. Existing approaches moderate AI output behaviourally, leaving open what the explanation layer received, where its output travelled, and what it was allowed to claim. We present Banking Shield, a fictional, non-bank, research-only banking-adjacent prototype whose central design inversion is to treat bounded *evidence of absence* — gate-backed evidence that sensitive values were not recorded in the frozen evidence pack and did not reach the explanation payload — as the deliverable. Banking Shield combines a fail-closed metadata firewall whose rejections are themselves audit evidence, per-session tamper-evident HMAC audit

chains with participant withdrawal semantics, and a deterministic, offline AI-style privacy firewall: an allowlist input contract, a deterministic mock narrative provider (no live LLM), a negation-aware output claim firewall, and a per-response privacy receipt, with a static source gate checking the four AI-firewall modules for network primitives. At the evidence freeze, all automated gates passed (417/417 unit tests; 43/43 end-to-end checks; 27/27 security checks; three privacy audits), and a formative dry run with five trusted internal testers completed 30 sessions with zero sensitive values in the evidence and 5/5 internal-test checklist comprehension of the system’s non-claims. The prototype is research-only and single-node; we state its limitations explicitly and argue that structural, fail-closed enforcement with machine-checked claim discipline is a reusable pattern for AI-assisted explanation in privacy-sensitive settings.

1 Introduction

Banking Shield is a *fictional* banking-adjacent research prototype: it has no connection to any bank, processes no real financial data, and makes no fraud-detection, scam-prevention, payment-safety, payee-verification, financial-advice, or compliance claims. We state this before anything else because the project’s central contribution is precisely the machinery that keeps such claims — and such data — out of the system.

Prototypes in sensitive domains usually argue safety by *policy* (“we do not collect X”) and argue explanation quality by *behaviour* (“the model rarely says Y”). Three gaps follow:

- **G1 — Explanation layers leak by default.** An LLM-backed explanation feature typically receives the session record and emits free text; nothing structural prevents sensitive input or unbounded output.
- **G2 — Overclaims are unpoliced.** Nothing stops an explanation layer from asserting capabilities (“fraud detected”) that the system does not have; in regulated-adjacent domains this is the costliest failure mode.
- **G3 — Absence of data is rarely evidenced.** Systems log what they did, not what they refused; a participant cannot verify that nothing sensitive was recorded.

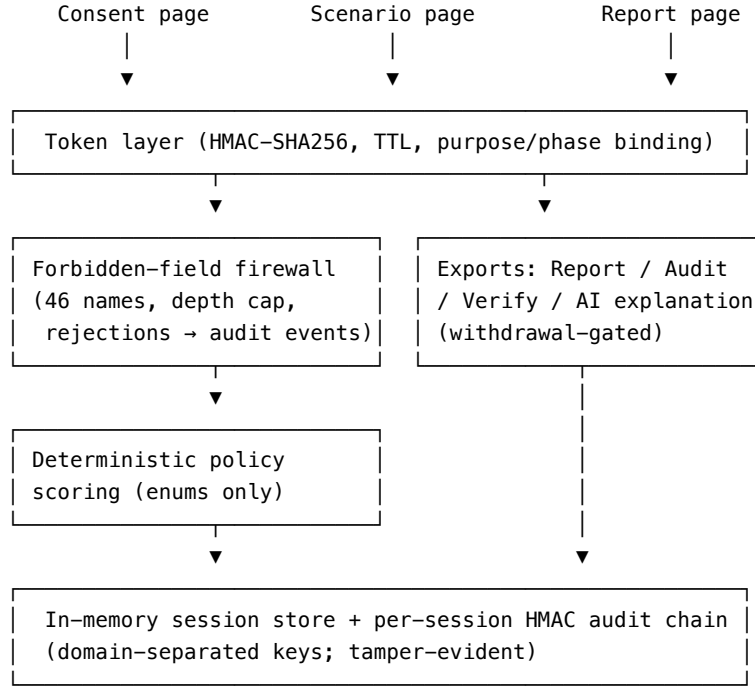
Our thesis: **structural, fail-closed metadata firewalls with machine-enforced claim discipline are better than content-level filtering for adding AI-style explanations to privacy-sensitive research prototypes.** Contributions:

- **C1** A metadata-only integrity-evidence architecture: a recursive forbidden-field firewall (46 key names, including prototype-pollution keys) whose rejections are audit events that escalate subsequent policy scoring — refusals become evidence (§4.1).
- **C2** A deterministic, offline AI privacy firewall: allowlist input contract, deterministic narrative provider, fail-closed output claim firewall, per-response privacy receipt, and a static source no-egress check over the four

AI-firewall modules with a negative self-test (§4.2, Figure 3).

- **C3** Claim discipline as a mechanism: a negation-aware scanner that blocks affirmative capability phrasing while passing required disclaimers, enforced at runtime and mirrored by repository CI (§4.3).
- **C4** A verifiable participation lifecycle — consent, withdrawal, HMAC audit chains, aggregate-only human evidence — exercised in a formative internal dry run (§4.4, §6.2).

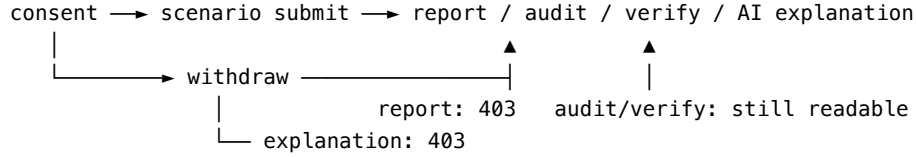
Figure 1 — Banking Shield architecture. *Every write passes the forbidden-field firewall before validation; every export is token-bound and chained. The system is metadata-only by construction (C1).*



2 Background and Motivation

A Banking Shield session walks a participant through consent, one of five fictional scenarios (a mock data-sharing consent screen, a mock payee-name check, a remote-access caution, a payment-pause prompt, and an AI-agent finance approval), and an export page offering a Report, an Audit listing, a Verify check, and — behind a default-off flag — an AI-style explanation (Figure 2). All scoring is a deterministic local function of enumerated metadata; the “policy outcome” (safe/warning/critical) is a fictional prototype output, not a calibrated alert.

Figure 2 — Session lifecycle. *Withdrawal blocks report and explanation export (403) but preserves audit/verify reads for transparency (C4).*



Three observations from the project’s own evidence motivate the design:

- **O1.** Every class of sensitive input the project could imagine (46 field names: credentials, OTPs, account identifiers, amounts, payees, device telemetry, structural pollution keys) appeared *only* in attack-style tests — and each was rejected and logged, suggesting refusal-as-evidence is mechanically achievable.
- **O2.** In the human dry run, all five testers understood the non-claims (5/5 on fictional-only, no-bank-connection, no-fraud-detection, no-financial-advice, withdrawal), but only one of five initially understood the Report/Audit/Verify exports — interpretability, not privacy, was the weak point, motivating an explanation layer. This echoes the broader finding that warning comprehension depends strongly on text characteristics and terminology (Lindell & Perry 2011, doi:10.1111/j.1539-6924.2011.01647.x; Scherr et al. 2015, doi:10.1002/acp.3195).
- **O3.** The riskiest place to add that explanation layer is exactly where G1/G2 bite; hence the firewall-first design of §4.2.

Table 1 — Allowed vs forbidden data (T1).

Allowed into the narrative payload (allowlist)	Forbidden everywhere (46-name denylist, classes)
Hashed session id	Credentials and one-time codes
Scenario type (enum)	Account/card/payment identifiers
User action/decision category (enum)	Amounts, balances, payees, references
Policy outcome + score (deterministic)	Transaction/statement content
Manual-review flag (boolean)	Screens, recordings, keystrokes, clipboard
Privacy-assertion booleans	Window titles, process/app names, device ids
—	__proto__, prototype, constructor

3 Threat Model

Assets. A1 sensitive banking payloads (must never enter); A2 audit-chain integrity; A3 the claim boundary; A4 participant anonymity; A5 official policy-result integrity.

Table 2 — Threats and controls (T5).

Threat	Controls	Asset
T1 Sensitive/polluted submission	Recursive denylist firewall, depth cap, body cap, rejection audit events, score escalation	A1
T2 Provider exfiltration (future live LLM)	Allowlist input (hashed id, enums), 4 KB cap, static no-egress gate, deterministic mock provider	A1
T3 Capability overclaim in narrative	Output schema allowlist, length caps, negation-aware claim scan, fail-closed rejection	A3
T4 Narrative drifts official result	Equality check against the authoritative record	A5
T5 Audit-log tampering	Per-session HMAC chains, domain-separated keys; tampered fixtures fail verification	A2
T6 Token theft/replay	HMAC tokens, constant-time compare, TTL, purpose/phase binding, path/token match	A1,A4
T7 Withdrawal bypass	Withdrawn $\rightarrow$ 403 on report/explanation; audit/verify intentionally readable	A4

Trust assumptions and residual risks. The server, its secrets, and transport security are trusted; Phase B testers are trusted insiders. The audit chain is tamper-*evident*, not tamper-proof: the server holds the keys — a limitation shared by server-side secure-logging schemes generally, which motivates third-party or distributed verification in that literature (Sree & Bhanu 2019, doi:10.1002/cpe.5143; Ali et al. 2021, doi:10.1002/ett.4272). Denylist scanning is incomplete by construction. Receipts attest process, not ground truth. Withdrawal does not erase the metadata-only audit chain — by design and disclosed. Out of scope: DoS beyond rate limiting, side channels, browser compromise, a malicious operator.

4 Design

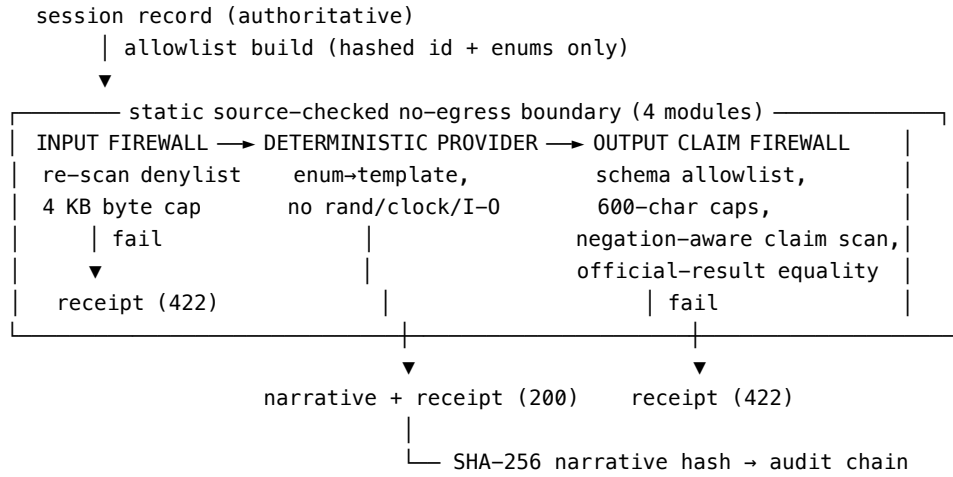
4.1 Metadata-only integrity evidence (C1)

Every write is filtered by a recursive forbidden-field firewall before validation: 46 forbidden key names spanning credentials, financial identifiers, content fields, device telemetry, and the structural pollution keys `__proto__/prototype/constructor`, with a recursion depth cap and a 16 KB body limit. A rejected attempt is not silently dropped: it appends a rejected-attempt audit event (route, reason, field name — never the value) and escalates the session’s subsequent policy scoring. This operationalises the data-minimisation principle — collect only what the stated purpose requires (Klein et al. 2021, doi:10.1111/1911-3846.12735; Yeung & Bygrave 2021, doi:10.1111/rego.12401) — as an enforced, evidence-producing mechanism rather than a policy statement; no compliance property is claimed or implied.

Alternative considered: schema-level allowlisting alone — rejected because an allowlist validates shape but does not produce evidence of hostile attempts; the denylist layer converts attacks into auditable signal. Both are used.

4.2 The deterministic AI privacy firewall (C2)

Figure 3 — The AI privacy firewall (page-one figure candidate). *The no-egress boundary is source-checked over the four AI-firewall modules; every stage fails closed to a receipt, never to a narrative (C2).*



The explanation pipeline has four fail-closed stages:

1. **Input firewall.** The narrative payload is *constructed*, not passed through: a hashed session id, scenario/action enums, the policy outcome, and the privacy-assertion booleans. The assembled payload is re-scanned against the forbidden-field denylist (defence in depth) and capped at 4 KB. Failure returns a receipt, never a narrative.
2. **Deterministic provider.** Narrative text is an enum→template pure function: no randomness, no clock, no I/O. A static source gate checks that the four AI-firewall modules import no network primitive, and a negative self-test confirms the gate fails when a network primitive is injected. This is not a host-level egress-control claim.
3. **Output claim firewall.** The narrative must match an exact top-level schema (field allowlist), respect 600-character caps per field, contain a non-empty non-claims list, and pass a negation-aware scan over 28 affirmative-capability phrasings; an equality check proves the narrative did not drift the official result.
4. **Privacy receipt.** Every response — success, blocked, or disabled — carries a receipt: provider identity (`deterministic_mock`), `sensitive_payload_sent_to_ai:false`, `network_egress_used:false`, official-result and claim-guard flags, and a success-only SHA-256 narrative hash

that is also written into the session’s audit chain.

Alternative considered: a live LLM behind the same firewall. Rejected for this work: it would convert the no-egress property from statically provable to operationally asserted. **What survives a future live provider:** the input allowlist, the byte cap, the output schema and claim scan, and the receipt mechanism are provider-independent; what is lost is determinism and reproducible narrative hashing, which would need replay logging instead. The mock therefore establishes the *contract and enforcement points*, which is the claim we make — not that a live model has been filtered or validated.

4.3 Claim discipline as a mechanism (C3)

The output firewall’s scanner is negation-aware: required disclaimers (“not fraud detection”, “not a fraud detection tool” — one determiner permitted after the negator) pass, while affirmative phrasing and weakened negation (“not really a ...”) are blocked. The bias is fail-closed: a false positive blocks a narrative; no false negative class is claimed. The same boundary is enforced a second time in repository CI by an overclaim-wording scanner — which, notably, flagged the firewall’s own denylist during integration (the denylist literally contains the phrases it blocks) and was resolved by an explicit scanner exclusion rather than string obfuscation. Claim discipline is thus testable at two independent layers.

4.4 Verifiable participation lifecycle (C4)

Each session carries an HMAC-SHA256 audit chain with domain-separated keys (participant-code hashing and chain signing cannot be cross-validated). Participants can export their Report (whose privacy assertions are all `false` — nothing sensitive recorded), the Audit listing, and a Verify check of chain consistency. Withdrawal blocks report and explanation export but intentionally preserves audit/verify reads. Human evidence is aggregate-only by construction.

Table 3 — Claims vs non-claims (T4).

The paper claims	The paper does NOT claim
Metadata-only integrity evidence (about the process)	Fraud detection or scam prevention
A fictional banking-adjacent prototype	Real banking protection or payment safety
A trusted internal dry run (formative)	Real payee verification or financial advice
A deterministic AI privacy firewall (contract + gates)	CDR/APRA/AML/CTF compliance
AI-style explanation without sensitive payloads	Production readiness or bank-grade security

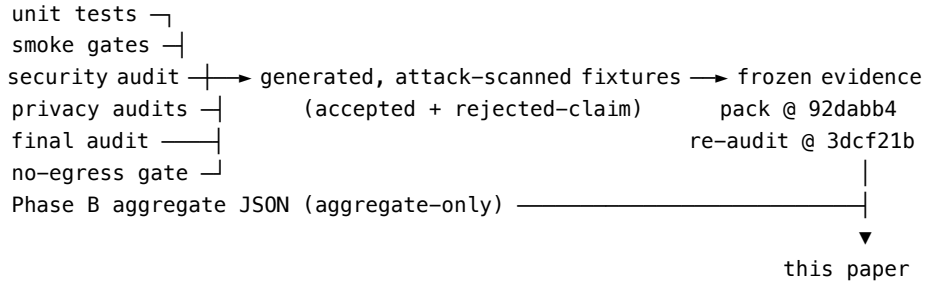
5 Implementation

Node.js/Express; static HTML/CSS/JS tester pages with `textContent`-only DOM writes in the explanation panel; an in-memory session store; shell/Node CI gates. The explanation endpoint is behind a default-off flag requiring the exact string `"true"`, is token-bound, path-matched, and read-rate-limited. The prototype is single-node and research-only.

6 Evaluation

We evaluate three questions: does the structural machinery hold under its own gates (E1); can trusted humans complete and comprehend the lifecycle (E2); and does the claim firewall block what it must while passing what it must (E3)? Figure 4 shows how evidence flows from gates to this paper.

Figure 4 — Evidence pipeline. *Every claim in this paper traces to a gate, fixture, or frozen aggregate (E1–E3).*



6.1 Gate results (E1)

Hypothesis: all structural properties are mechanically checkable. *Result:* at the evidence freeze, every automated gate passed (Table 4). (Table 4: every gate green at freeze — the claim is about gates, not about the world.)

Table 4 — Gate results at evidence freeze (T3).

Gate	Result
Unit tests	417/417 pass
Banking smoke	14/14 pass
AI firewall smoke	5/5 pass
Full banking E2E smoke	43/43 pass (incl. tamper-evidence check)
Banking security audit	27/27 pass
Banking privacy audits	all 3 PASS
No-egress static gate	PASS (with negative self-test)
Dependency audit	0 vulnerabilities
Repository CI quality gate	green

6.2 Formative internal dry run (E2, exploratory)

Hypothesis: trusted insiders can complete the lifecycle and understand the non-claim checklist. *Result:* five trusted internal testers completed 30 sessions (Table 5). Zero real banking values were entered; zero sensitive values were found in evidence; the deterministic policy pattern reproduced identically across testers (safe/warning/warning/warning/safe across the five scenario types); all five withdrawals blocked subsequent report export; and comprehension of all five non-claim checks was 5/5. Export interpretability was the weak point (1/5 pre copy-patch); a focused 3-session rerun confirmed the revised copy but **cannot establish improvement** — we report it as a design iteration. This is a formative dry run with n=5 trusted insiders, not a user study and not a representative banking-customer sample; no generalisation is claimed. (*Table 5: the dry run evidences lifecycle completion and non-claim-checklist comprehension, nothing more.*)

Table 5 — Phase B aggregates (T2, exploratory).

Measure	Value
Trusted internal testers	5
Total sessions	30
Submitted scenario sessions	25 (5 per scenario)
Withdrawal sessions	5
Withdrawals blocking later report export	5/5
Real banking values entered	0
Sensitive values found in evidence	0
Forbidden payload structures in evidence	0
Comprehension: fictional-only	5/5
Comprehension: no bank connection	5/5
Comprehension: no fraud detection	5/5
Comprehension: no financial advice	5/5
Comprehension: withdrawal	5/5
Export interpretability (pre copy-patch)	1/5 (design iteration followed)

6.3 Fixture studies (E3)

Hypothesis: the output firewall passes a clean narrative and blocks a poisoned one. *Result:* the accepted-explanation fixture (clean generated narrative with receipt) passes all gates and contains no attack values; the rejected-claim fixture — a deliberately poisoned narrative asserting a detection capability — is blocked by the claim guard with the failed gate recorded in its receipt. (*Fixture pair: one accepted, one rejected, both attack-value-scanned — the firewall’s two required behaviours, demonstrated.*)

7 Limitations

- (1) The narrative provider is a deterministic mock; no live LLM has been filtered or validated. (2) n=5 trusted insiders; no statistical claims, no representative-user claims, and no banking-customer comprehension claims.
- (2) Denylist claim scanning is incomplete by construction; adversarial phrasing can evade it, and the allowlist construction is the stronger control. (4) The gates are written by the project they validate; mitigated by negative self-tests and public reproducibility, not eliminated. (5) Single-node, in-memory prototype. (6) The audit chain is server-keyed: tamper-evident to participants, not tamper-proof against the operator. (7) The no-egress claim is a static source check over the four AI-firewall modules, not a network sandbox or host-level egress-control guarantee.

8 Reproducibility and Review-Substitute Pack

The artifact includes the paper, evidence pack, final audit, claim audit, and Stage B5-R review-substitute pack under `docs/research/banking-pilot/paper/`. The Stage B5-R pack simulates five external-style hostile reviews (privacy/security, banking governance, HCI, AI safety, and reject-oriented reviewer #2 attack) and records author responses before preprint submission. This is not a replacement for formal peer review; it is a structured preprint-risk-reduction step.

The core reproduction gates are:

```
npm test
bash scripts/smoke-banking-pilot.sh
SIMURGH_BANKING_PILOT_AI_EXPLAIN=true bash scripts/smoke-banking-pilot-ai-firewall.sh
bash scripts/smoke-banking-pilot-full-e2e.sh
bash scripts/security-audit-banking-pilot.sh
node scripts/privacy-audit-banking-pilot.mjs
node scripts/privacy-audit-banking-pilot-phase-b.mjs
node scripts/privacy-audit-banking-pilot-ai-firewall.mjs
npm audit --audit-level=moderate
```

These gates support the paper’s mechanism and evidence-pack claims. They do not establish production security, regulatory compliance, live-LLM safety, or real-world banking effectiveness.

9 Related Work

Grouped by approach. Academic citations below were verified and carry DOIs; open-banking and payment-scheme citations use stable institutional sources.

Tamper-evident logging. Secure-logging schemes protect log integrity and authenticity with hash chains, MACs, and related structures, and survey the

limits of server-trusted designs (Sree & Bhanu 2019, doi:10.1002/cpe.5143; Ali et al. 2021, doi:10.1002/ett.4272). We apply standard HMAC-chain machinery per-session, with participant-facing verify exports and withdrawal semantics; our chain is deliberately scoped as tamper-evident, not tamper-proof.

Warning and rights comprehension. Warning research shows comprehension hinges on reception, attention, and terminology (Lindell & Perry 2011, doi:10.1111/j.1539-6924.2011.01647.x), that recipients frequently take no action on textual alerts (Kim et al. 2019, doi:10.1111/1468-5973.12278), and that text characteristics measurably drive comprehension of rights-style warnings (Scherr et al. 2015, doi:10.1002/acp.3195). Banking Shield measures comprehension of *non-claims* — what the system does NOT do — an inverted warning problem, in a formative setting only.

Data minimisation and design-based regulation. The GDPR’s data-minimisation and data-protection-by-design principles require systems to hold only what their purpose strictly needs and to hard-wire protection into architecture (Klein et al. 2021, doi:10.1111/1911-3846.12735; Yeung & Bygrave 2021, doi:10.1111/reg.12401; Bradshaw & DeNardis 2019, doi:10.1002/poi3.195). Our firewall is one concrete design-based mechanism, with the additional property that refusals produce evidence; we claim no compliance status under any regulation.

LLM guardrails and AI risk frameworks. Guardrail practice moderates live model output with content filters, usage monitoring, and evaluation frameworks (Lee et al. 2024, doi:10.1111/bjet.13505), and AI trust/risk/security frameworks catalogue runtime enforcement and output filtering controls (Ray 2026, doi:10.1111/exsy.70213). These filter content from a live model; Banking Shield instead makes the provider structurally unable to receive sensitive input, statically unable to egress, and schema-bound on output, with content scanning only as the final layer.

Open-banking consent UX. Open-banking customer-experience guidance treats consent and access dashboards as critical user-control surfaces (Open Banking Implementation Entity n.d.; World Bank n.d.). Banking Shield does not evaluate real open-banking consent efficacy; it evidences the integrity of a fictional consent-style flow.

Payee-confirmation services. Confirmation-of-payee and verification-of-payee schemes perform account-name matching before payment initiation (Pay.UK n.d.; European Payments Council n.d.). Banking Shield’s payee scenario is fictional: the contribution is the metadata boundary, not name matching.

Per-response AI transparency artifacts. Model cards and datasheets document models or datasets as release-time artifacts (Mitchell et al. 2019, doi:10.1145/3287560.3287596; Gebru et al. 2021, doi:10.1145/3458723). Banking Shield’s receipts are narrower: per-response, machine-checked records of what the explanation pipeline did and did not receive.

10 Conclusion

We asked whether an AI-style explanation layer can be added to a privacy-sensitive research prototype without inheriting the leak and overclaim risks of live generative pipelines. Banking Shield answers with a structural pattern: fail-closed metadata firewalls, a deterministic offline provider behind provider-independent enforcement points, machine-checked claim discipline, and per-response receipts anchored in tamper-evident audit chains. At the evidence freeze every automated gate passed and a formative five-tester dry run completed with zero sensitive values recorded and full non-claim-checklist comprehension — evidence that, within its research-only scope, the mechanism remained intact under the stated gates and formative dry-run scope. Future work: replay-logged live-provider integration behind the same contract, allowlist-template generation as a stronger guarantee than denylist scanning, externally countersigned receipts, external security review, and a powered user study.

11 LLM-Assistance Disclosure

This paper was drafted with substantial assistance from a frontier language model operating under a logged, evidence-pack-constrained protocol (`docs/research/banking-pilot/stage-b5-model-paper/`): the model synthesised, critiqued, and drafted; it validated nothing. A second model-assisted independent-style review stage (`docs/research/banking-pilot/paper/review/`) simulated five hostile reviewer roles and produced an author response before this v1.1 preprint candidate. All factual statements trace to automated gate results, generated fixtures, frozen aggregate evidence, and explicit audit records. Citation candidates were verified before inclusion; unresolved placeholders are not retained in this version.

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